MATHEMATICS

The Honest Coder

Maths trick #9

Website → https://thehonestcoder-edu.github.io/

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Trick: Squaring a 2-Digit Number

Step by Step Explanation
(with examples)

d1 = first digit of the number
d2 = second digit of the number
Thus, the number is (d1 d2).

$$\frac{(d1\ d2)^2}{d1^2 \mid 2*d1*d2 \mid d2^2}$$

First part = $d1^2$
Second part = 2 * d1 * d2
Third part = $d2^2$

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Arrange results from left to right.

Then, starting from the right, keep one digit, carry the rest, till the middle and last part has 1 digit. This is the answer.

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Example 1→

47^2

Here: d1=4, d2=7

$$4\ 7^2$$

16 | 2*4*7 | 49

First part = 16

Second part = 2*4*7 = 56

Third part = 49

Carry process:

Right → 49 → write 9, carry 4

Middle → 56 + 4 = 60 → write 0, carry 6

Left → 16 + 6 = 22

Final answer → 2204

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Example 2→

62^2

Here: d1=6, d2=2

$$\frac{6\ 2^2}{36 \mid 2*6*2 \mid 4}$$

First part = 36

Second part = 2*6*2 = 24

Third part = 4

Carry process:

Right → 4 → write 4

Middle → 24 → write 4, carry 2

Left → 36 + 2 = 38

Final answer → 3844

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Example 3→

93^2

Here: d1=9, d2=3

$$\frac{4\ 7^2}{16 \mid 2*4*7 \mid 49}$$

First part = 81

Second part = 2*9*3 = 54

Third part = 9

Carry process:

Right → 9 → write 9

Middle → 54 → write 4, carry 5

Left → 81 + 5 = 86

Final answer → 8649

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Practice sheet

1. 32^2
2. 47^2
3. 58^2
4. 73^2
5. 96^2

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(QR code → Website access)